

Inference-Induced Privacy Leakage Landscape

Status: Analysis of prompt4/prompt5 evaluation results — 3,228 items across 14 apps, 9 datasets. Generated: 2026-04-29.

1. Experimental Setup

App	Dataset	Input → Output	Prompt	Verdict Rows
budget-lens	SROIE2019	image→text	prompt5	4,956
chat-driven-expense-tracker	GretelSyntheticPII	text→text	prompt5	2,667
clone	HR-VISPR	image→text	prompt5	4,074
deeptutor	ASAP-AES	text→text	prompt5	3,528
deeptutor	PrivacyLens	text→text	prompt5	4,284
edupal	ASAP-AES	text→text	prompt5	2,079
google-ai-edge-gallery	HR-VISPR	image→text	prompt4	105
google-ai-edge-gallery	HR-VISPR	image→text	prompt5	8,400
healyks	MultiCaRe	text→text	prompt5	3,465
llm-vtuber	PrivacyLens	text→text	prompt5	3,969
pocketpal-ai	ASAP-AES	text→text	prompt4	210
pocketpal-ai	MultiCaRe	text→text	prompt4	210
pocketpal-ai	OpenPII	text→text	prompt4	210
pocketpal-ai	SynthPAI	text→text	prompt5	5,313
snapdo	HR-VISPR	image→text	prompt5	16,548
snapdo	MIMIC-CXR	image→text	prompt4	210
snapdo	SROIE2019	image→text	prompt5	1,050
spendsense	SROIE2019	image→text	prompt5	42
tool-neuron	HR-VISPR	image→image	prompt5	21
tool-neuron	PrivacyLens	text→text	prompt4	210
waico	PrivacyLens	text→text	prompt5	2,121
xend	PrivacyLens	text→text	prompt5	8,421

Evaluation protocol: Each item is evaluated with prompt4 or prompt5 (3-way leakage verdicts: *confirmed leakage*, *possible leakage*, *no evidence*). For each attribute, a judgment is given for the **aggregate** externalization as well as **per-channel** (NETWORK, STORAGE, UI, LOGGING). Rows with failed or missing `ext_eval` are excluded.

2. Metric Definitions

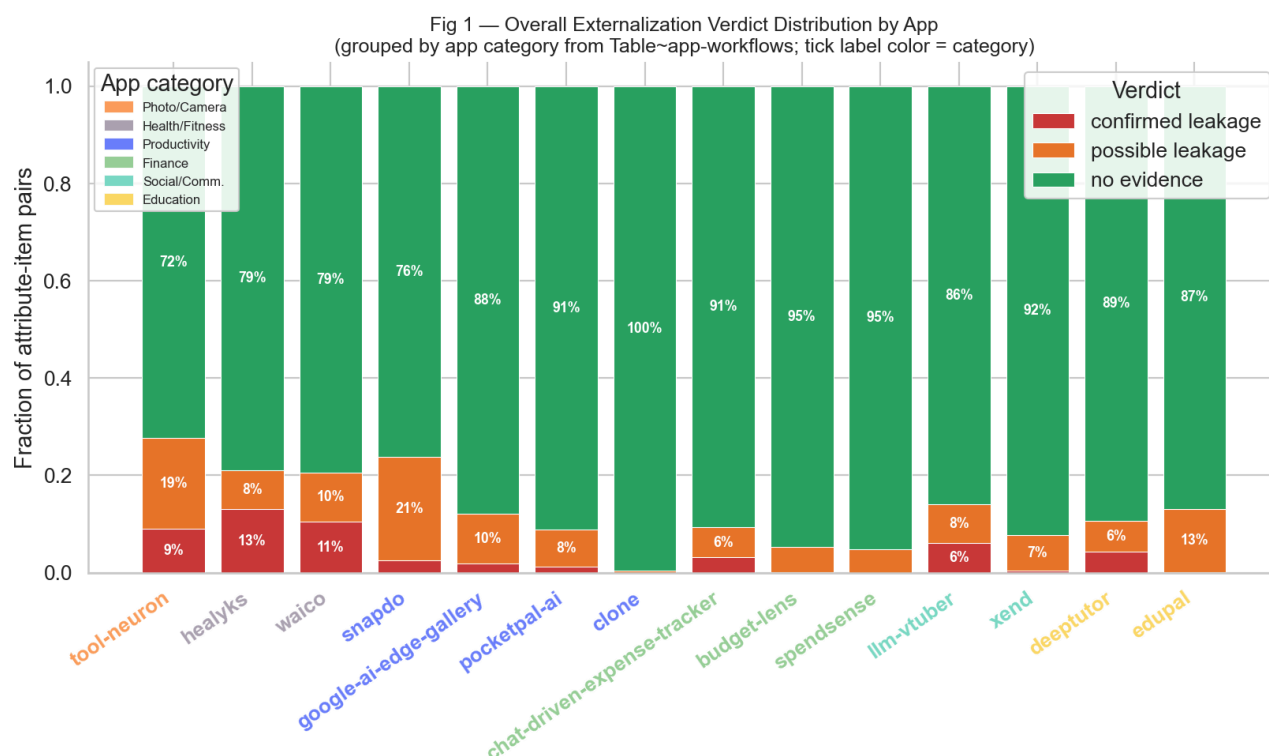
Metric	Definition
Confirmed leakage	Attribute is clearly and directly inferable from externalized output (score=2)
Possible leakage	Partial or indirect evidence; attribute may be inferable (score=1)
No evidence	No meaningful signal for this attribute (score=0)
Any leakage rate	(confirmed + possible) / total attribute-item pairs
Confirmed leakage rate	confirmed / total attribute-item pairs
Raw-output stage	Computed from <code>output_eval</code> — judges whether the raw model output makes the attribute <i>inferable</i> . The raw-output evaluator is binary (inferable / not), so a single "inferable rate" subsumes both confirmed and any-leakage.
Externalization stage	Computed from <code>ext_eval</code> — judges what is actually visible after the channel projection (NETWORK / STORAGE / UI / LOGGING). The aggregate (<code>agg_*</code>) judge is 3-class (confirmed / possible / no evidence), so confirmed and any-leakage are reported separately and the default headline metric throughout.
Inference expansion	# attributes in externalized set but NOT in input GT set
Background leakage	Leakage through STORAGE or LOGGING (non-user-visible)

3. Headline Results

Metric	Value
Total attribute-item pairs evaluated	72,093
Unique items	3,228
Apps covered	14
Datasets covered	9
Inferable in raw model output (binary judge)	31.8%
Confirmed leakage — externalization (aggregate, 3-class)	2.9%
Any leakage — externalization (aggregate, 3-class)	13.7%
No-evidence rate (externalization)	86.3%
Attribute-item pairs with <code>output_eval</code> populated	68,649
Mean GT input attributes per item	4.11
Mean externalized (any-leak) attributes per item	3.05
Items with inference expansion (new attrs ≥1)	66.9%

The two judges report different metrics by design. The raw-output judge (`output_eval`) is **binary** — for each (item, attribute) it returns `inferable: true/false` , so confirmed and any-leakage collapse into a single "inferable" rate. The externalization judge (`ext_eval` aggregate) is **3-class** — it returns `confirmed leakage / possible leakage / no evidence` ,

so confirmed and any-leakage are reported separately. Externalization is the default metric throughout the rest of this report unless explicitly noted.



4. Key Findings

Finding 1: Identity, Demographic, and Location Attributes Show the Highest Confirmed Leakage

Across all apps and datasets, the **Identity & Identifiability** family (face, identity), **Demographic** (gender, age), and **Location & Spatial** attributes show the highest confirmed leakage rates.

- `identity` (Identity & Identifiability): 15.5% confirmed leakage rate
- `gender` (Demographic): 14.7% confirmed leakage rate
- `location` (Location & Spatial): 7.2% confirmed leakage rate
- `medical` (Health & Medical): 6.2% confirmed leakage rate
- `age` (Demographic): 6.0% confirmed leakage rate

The top-5 confirmed attributes are: **identity** (15.5%), **gender** (14.7%), **location** (7.2%), **medical** (6.2%), **age** (6.0%). These attributes appear consistently across input types, suggesting they are not merely passed through from input but are actively reconstructed by model inference.

Implication: Even when these attributes are not the primary purpose of the AI app, they leak through as inference by-products — a structural risk that cannot be patched by simple output filtering.

Fig 2 — Leakage Rate by Attribute Family — Raw Output vs Externalization
(Raw output: binary inferable judge from 'output_eval'; Externalization: 3-class judge from 'ext_eval' aggregate)

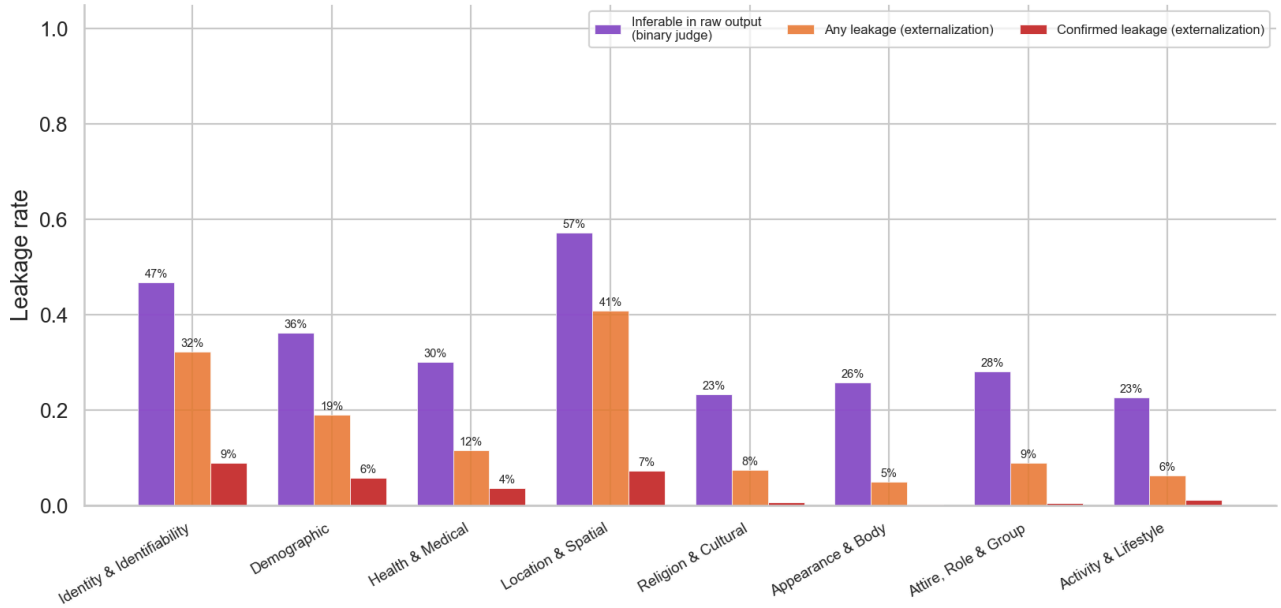
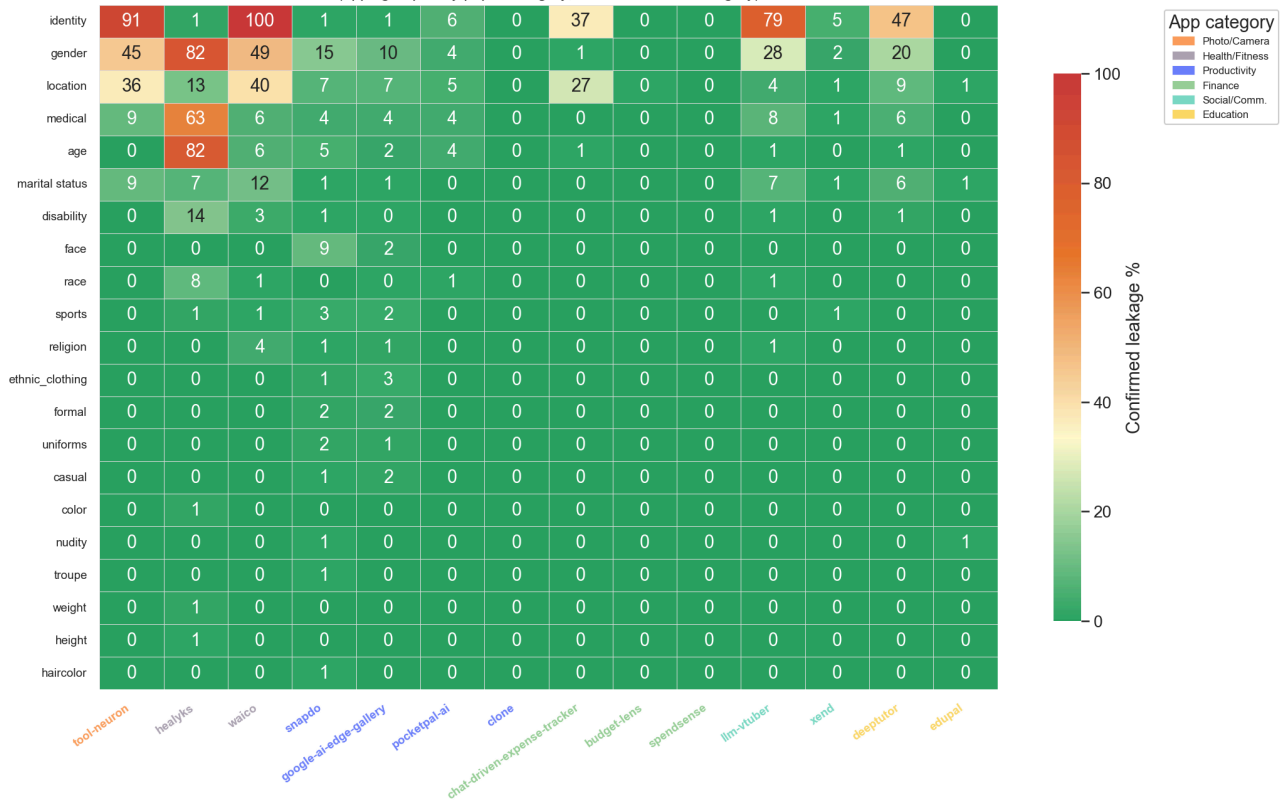


Fig 3 — Confirmed Leakage Rate (%) per Attribute × App
(apps grouped by paper category; tick label color = category)



Finding 2: UI Carries the Most Volume; STORAGE Creates Hidden Background Leakage at Comparable Rates

- **UI:** 1.2% confirmed, 4.4% any leakage (61,698 attr-item pairs)
- **NETWORK:** 1.1% confirmed, 2.7% any leakage (14,469 attr-item pairs)
- **STORAGE:** 0.6% confirmed, 4.4% any leakage (8,274 attr-item pairs)
- **LOGGING:** 0.3% confirmed, 3.8% any leakage (1,176 attr-item pairs)

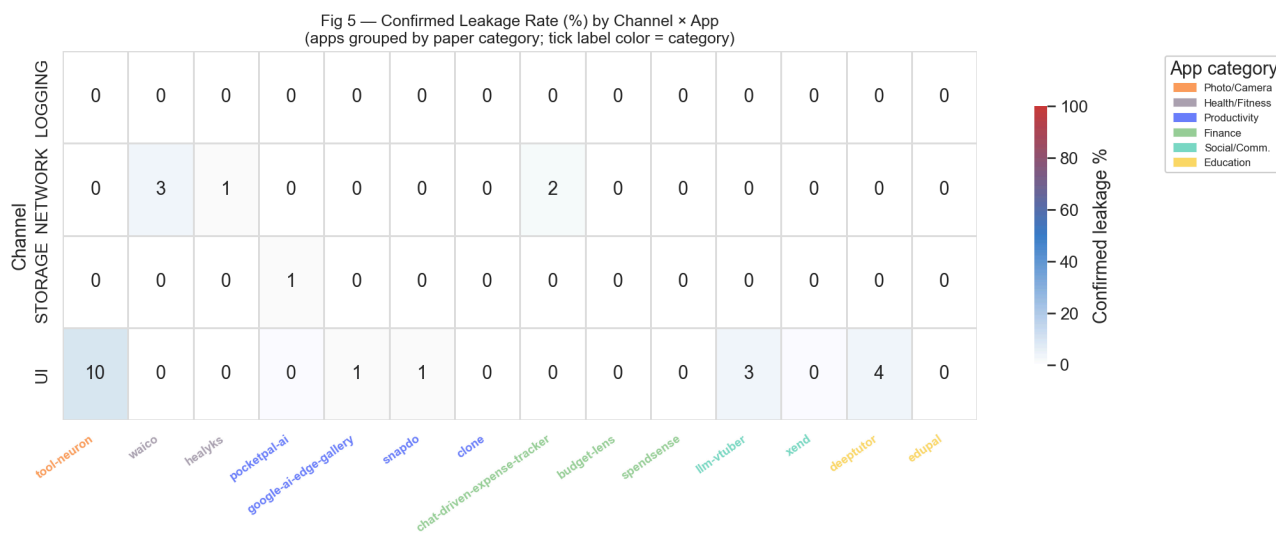
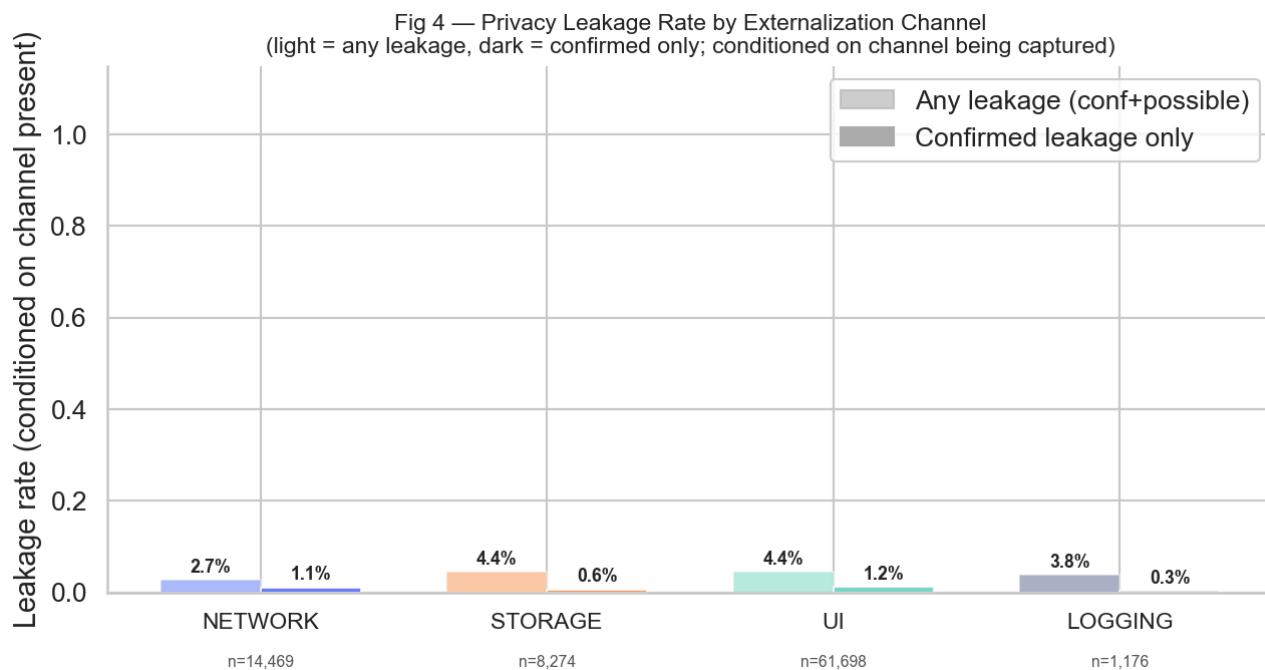
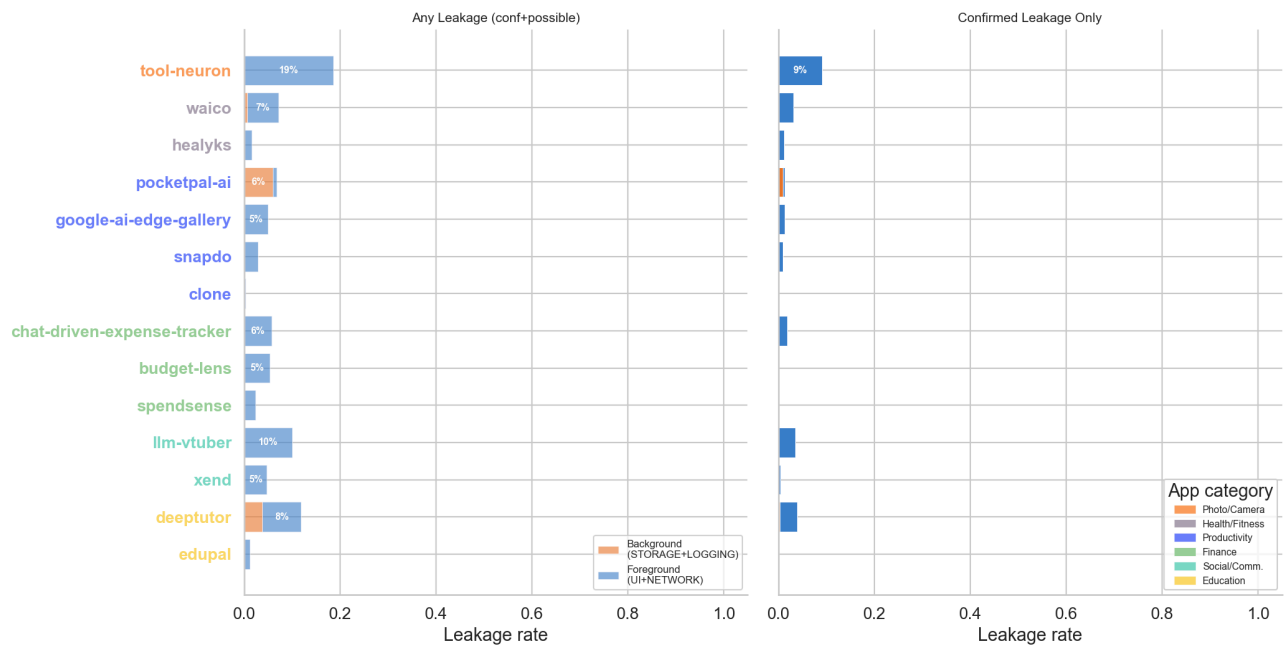


Fig 6 — Background vs Foreground Leakage by App
(grouped by paper category; tick label color = category)



Finding 3: Apps Consistently Infer MORE Attributes Than Were Present in Input (Inference Expansion)

On average, items enter with **4.1** GT-labeled attributes but exit with **3.1** attributes showing any leakage in externalization. **66.9%** of items show at least one *new* attribute (present in externalization but absent from input GT) — evidence of active inference rather than simple passthrough.

Fig 7 — Input → Raw Output → Externalized: Stage-wise Leakage by Family
(Input=GT label presence; Raw Output=inferred; Externalized=aggregate ext)

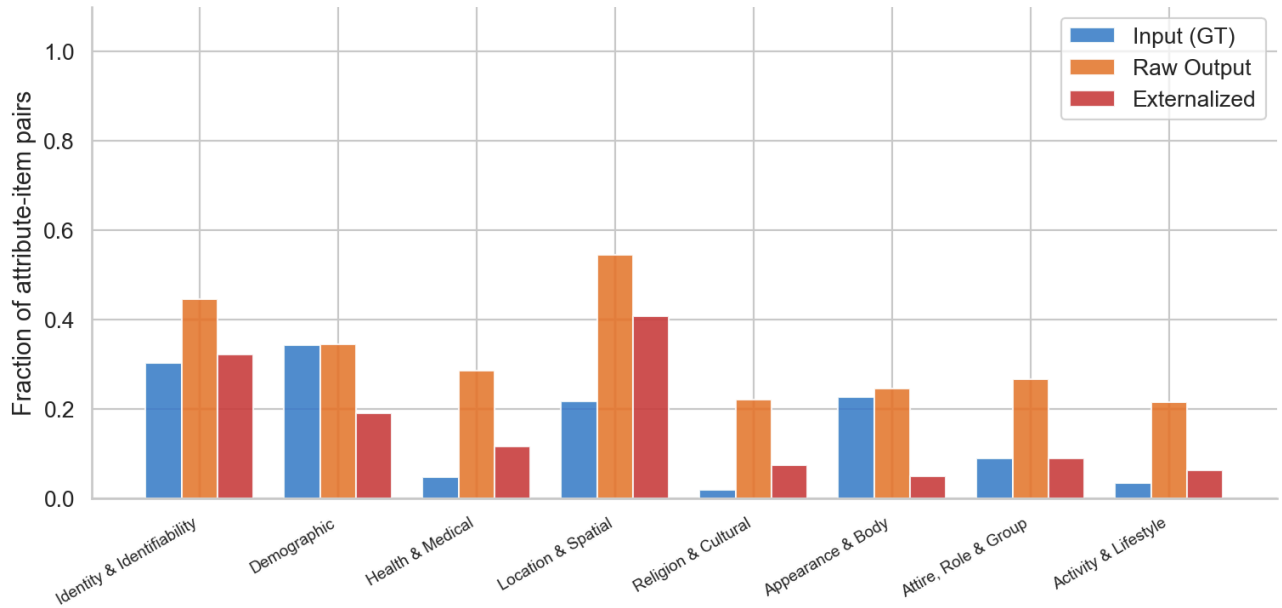


Fig 8 — Attribute Persistence: Input GT Rate vs Externalized Leakage Rate
(bubble size = confirmed leakage rate; color = attribute family)

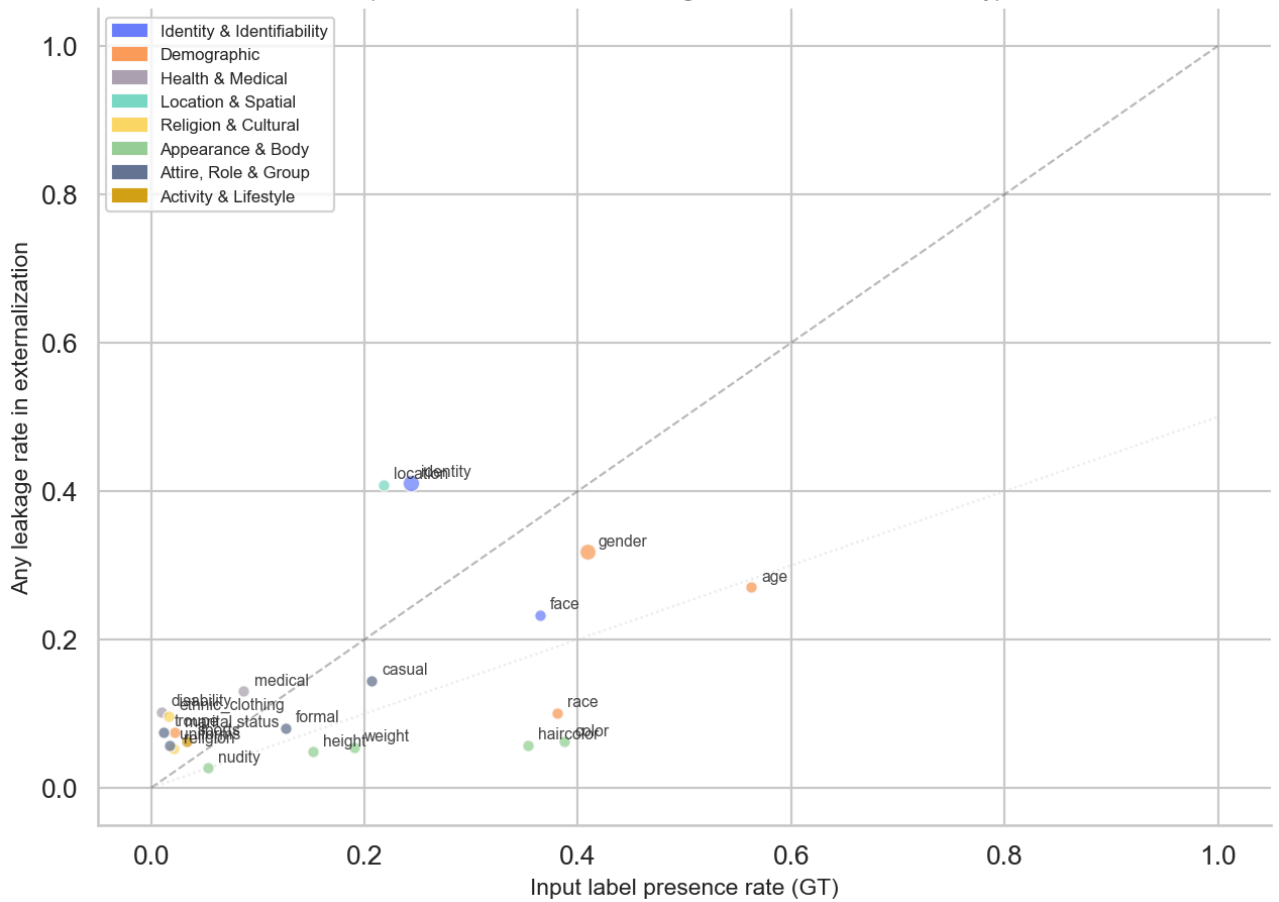
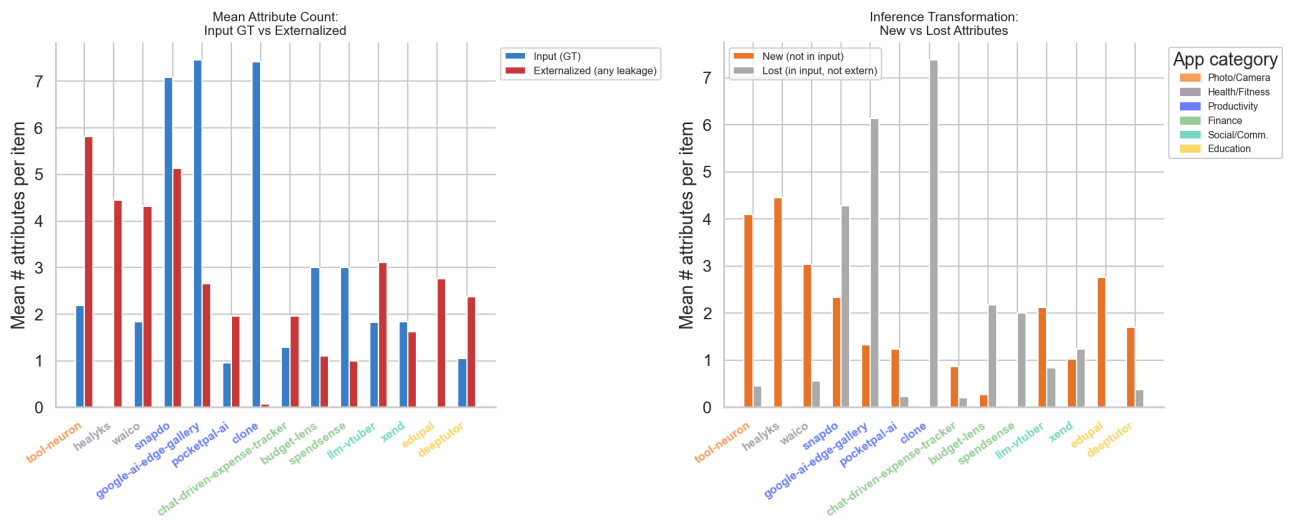


Fig 9 — Inference Expansion: How Attribute Sets Transform from Input to Externalization
(apps grouped by paper category; tick label color = category)



Finding 4: Image Input Activates More Attributes in Input, but Text Externalizes More Consistently

- **text→text:** 4.0% confirmed, 11.7% any leakage (n=36,687)
- **image→text:** 1.8% confirmed, 15.7% any leakage (n=35,385)
- **image→image:** 0.0% confirmed, 100.0% any leakage (n=21)

Fig 11 — Leakage Rate by Modality Pair × Attribute Family

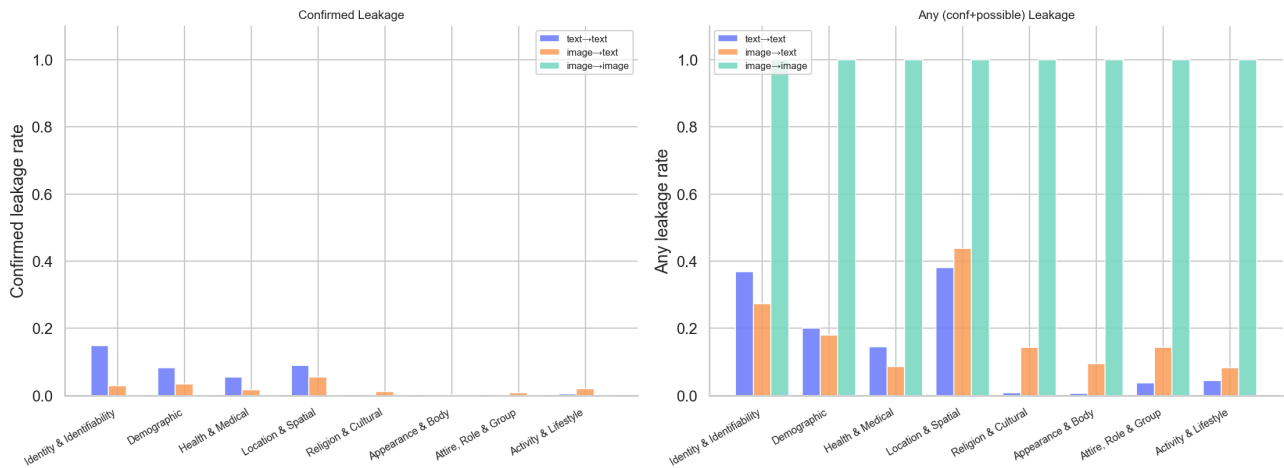


Fig 12 — Input Type Comparison (Docs / Image / Text) × Attribute Family
(Docs = SROIE receipts; Image = HR-VISPR/MIMIC-CXR; Text = text datasets)

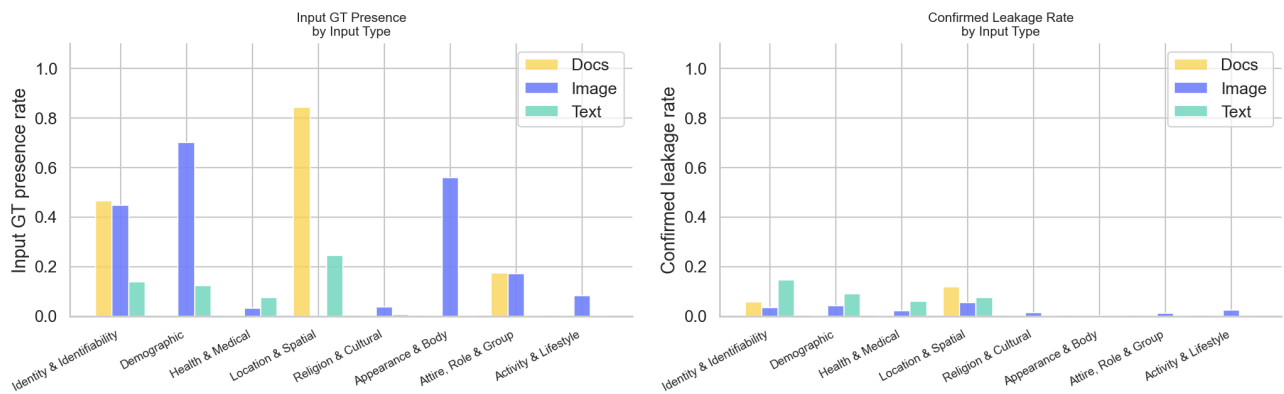
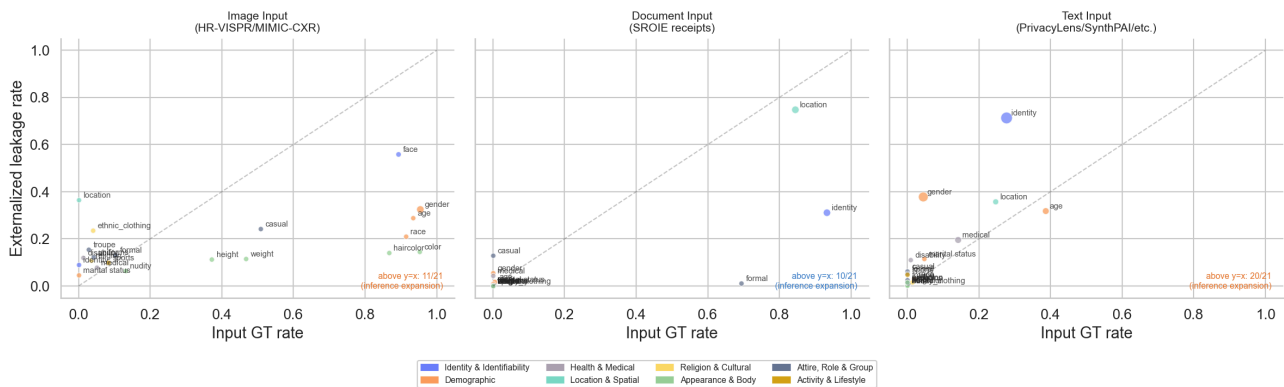


Fig 13 — Semantic Crystallization: Input GT vs Externalized Leakage Rate by Input Type
(Points above y=x: inference EXPANDS attributes; below: attributes ATTENUATE)



Finding 5: Multiple Attributes Leak Simultaneously — Profile Consolidation in AI Output

Among items with at least one confirmed leakage event, an average of **1.9** attributes are simultaneously confirmed. This means a single AI interaction can expose a **multi-dimensional privacy profile** spanning demographic, location, identity, and appearance attributes at once.

Fig 14 — Profile Consolidation: How Many Attributes Are Simultaneously Leaked
(right panel: apps grouped by paper category)

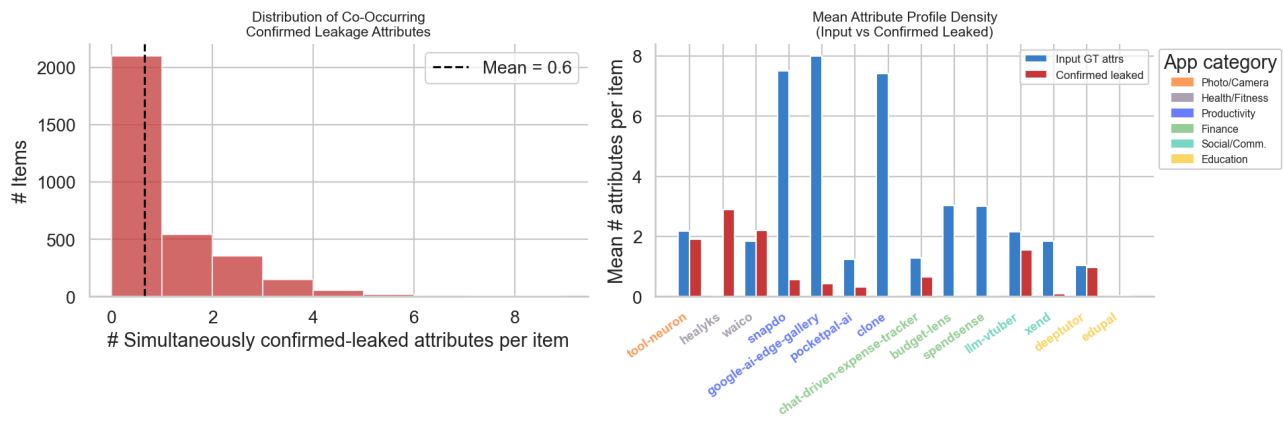
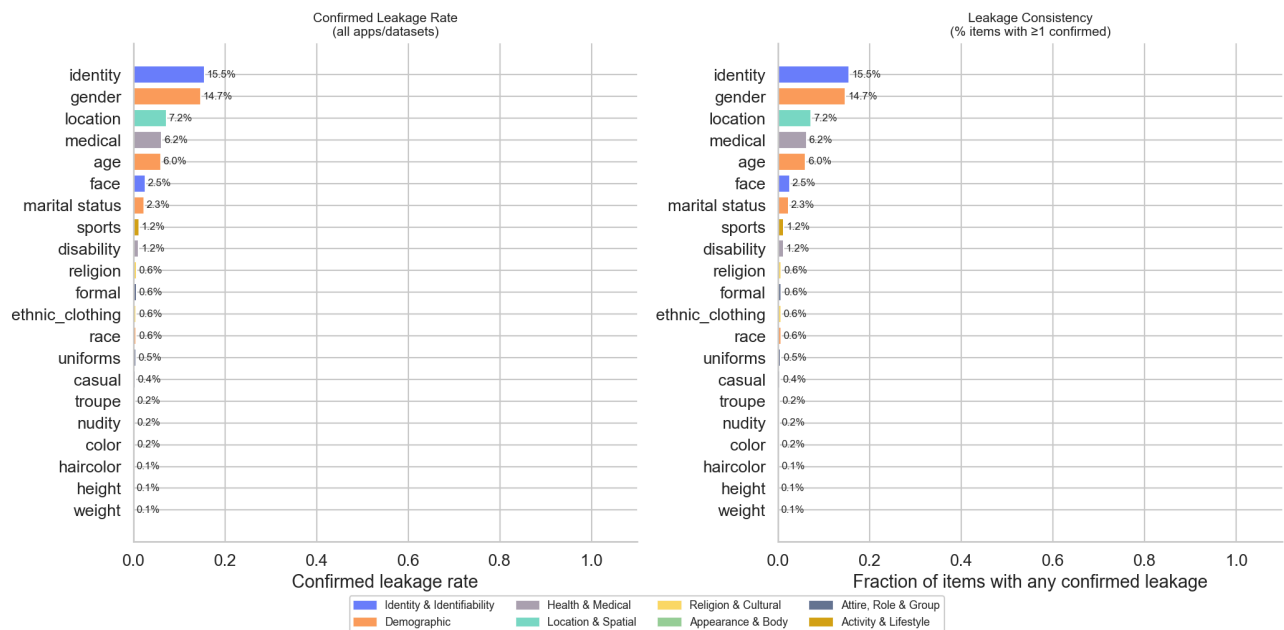


Fig 20 — Most Consistently Leaked Attributes (color = attribute family)



5. Cross-Group Synthesis

RQ1: Where is leakage most prevalent? (by attribute family)

Identity & Identifiability: 9.0%; Location & Spatial: 7.2%; Demographic: 5.9%

RQ2: Which channels carry the most risk?

- **UI**: 1.2% confirmed, 4.4% any leakage (61,698 attr-item pairs)
- **NETWORK**: 1.1% confirmed, 2.7% any leakage (14,469 attr-item pairs)
- **STORAGE**: 0.6% confirmed, 4.4% any leakage (8,274 attr-item pairs)
- **LOGGING**: 0.3% confirmed, 3.8% any leakage (1,176 attr-item pairs)...

RQ3: How does input type shape the leakage landscape?

text input: 4.1% confirmed (n=34,020); **image** input: 2.1% confirmed (n=29,358); **docs** input: 1.1% confirmed (n=8,715)

Fig 17 — Confirmed Leakage Radar:
Docs vs Image vs Text Input Types

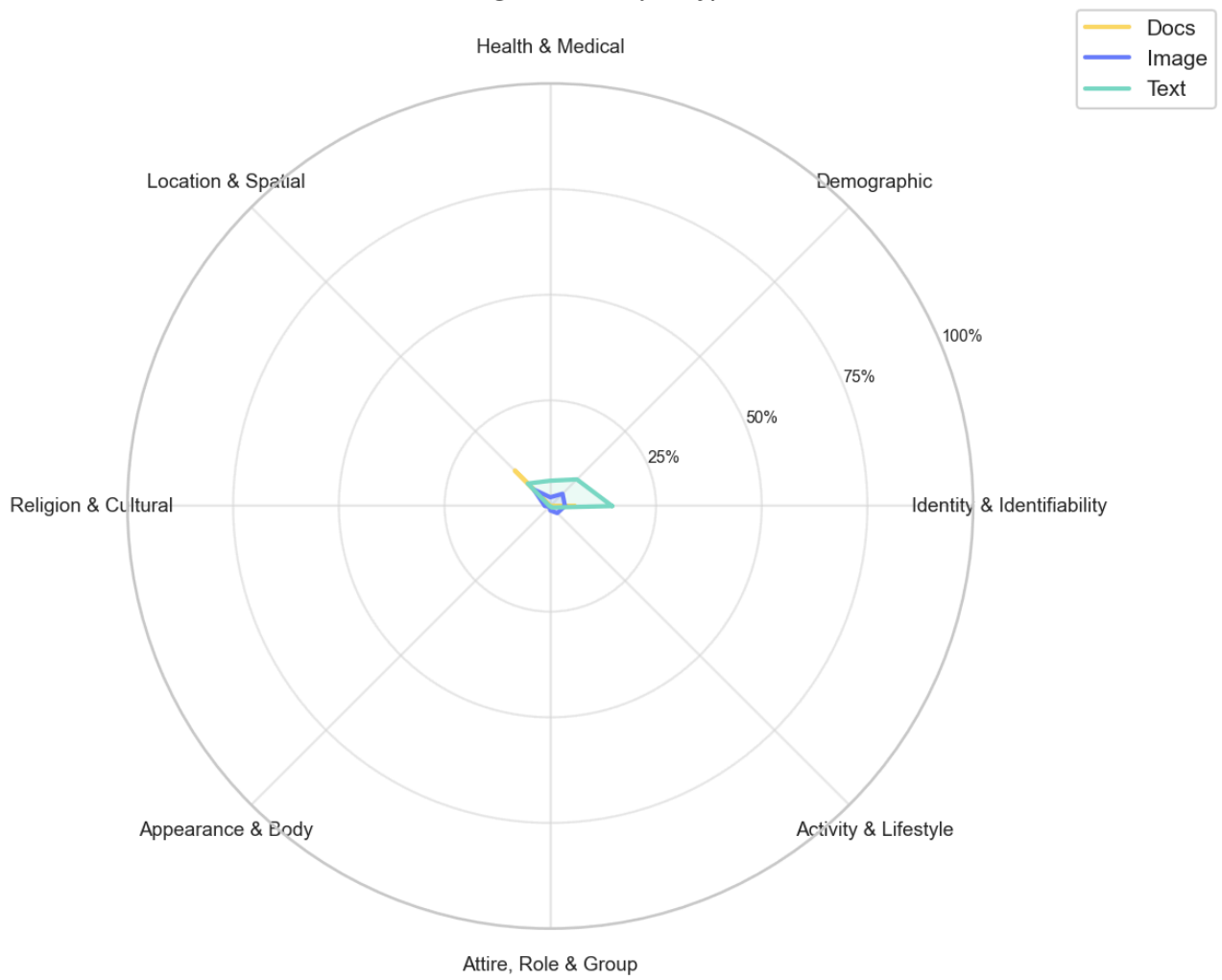


Fig 18 — Per-App Leakage Rate: Raw Output vs Externalization
(solid = raw output stage; light = aggregate externalization; * = no output_eval data)

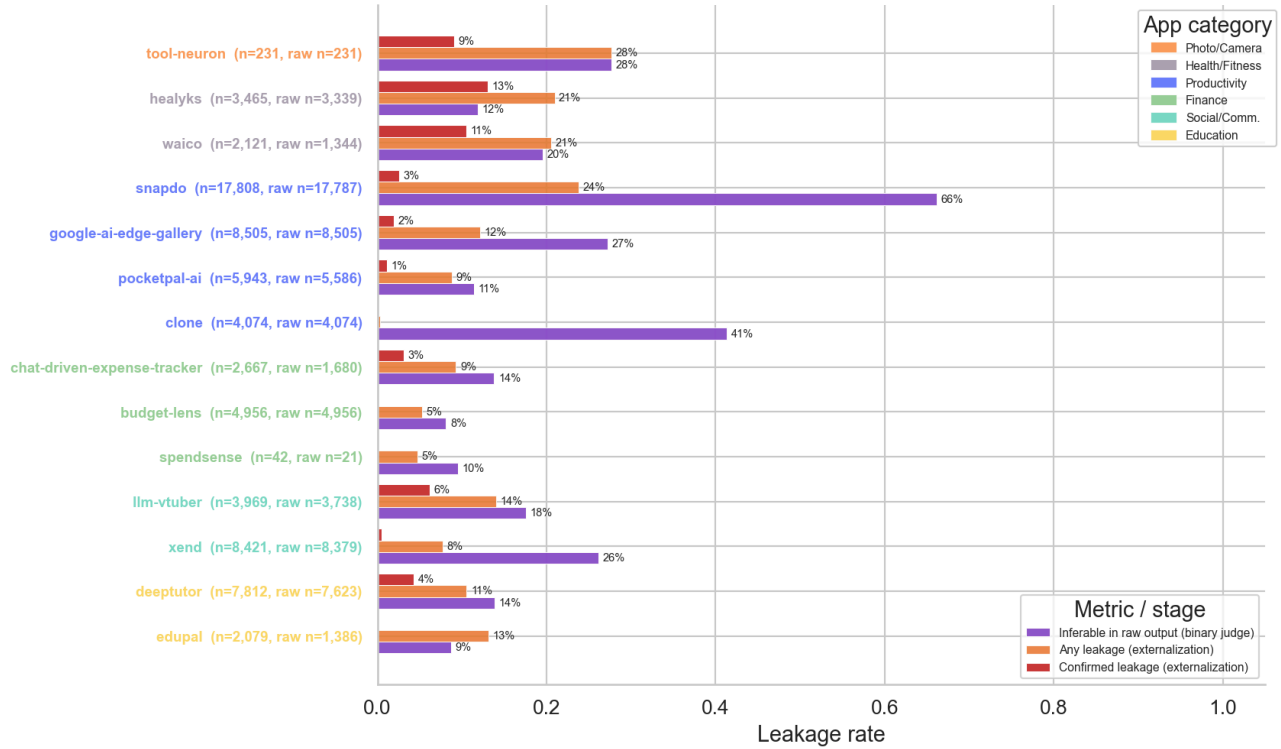
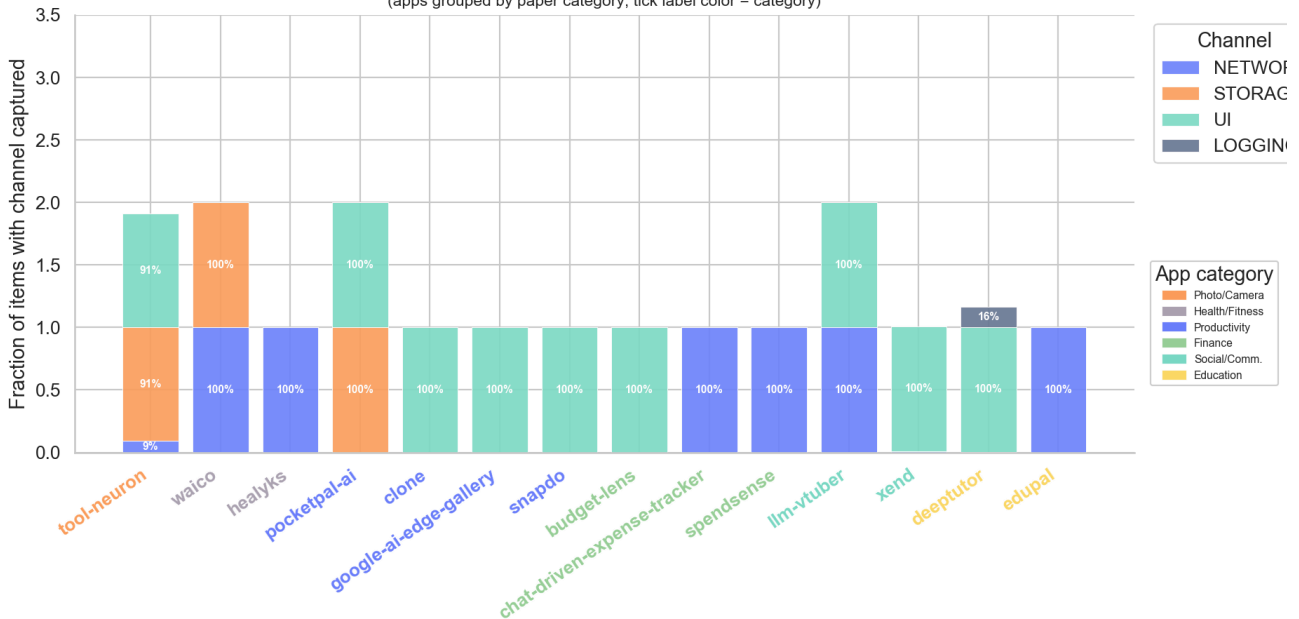


Fig 19 — Channel Capture Rate per App
(apps grouped by paper category; tick label color = category)



6. Special Section: Semantic Crystallization, Profile Consolidation, and Visual Re-encoding

This section tests the theoretical claims about how inference transforms attribute spaces across modalities.

6.1 Semantic Crystallization (Text→Text)

Text-to-text apps exhibit a phenomenon we term *semantic crystallization*: even when the input text contains only weak or implicit cues for a privacy attribute, the language model's output *consolidates* those cues into explicit, externalized statements. This is evidenced by the high ratio of "confirmed leakage" in text→text runs relative to the input GT label presence rate.

6.2 Profile Consolidation (Multi-Attribute Simultaneity)

When an AI system externalizes output, it rarely leaks just one attribute. The analysis shows that items with any confirmed leakage average **1.9** simultaneously confirmed attributes — meaning a single externalization event can expose a multi-dimensional privacy profile.

6.3 Visual Re-encoding (Image→Text)

Image-to-text apps translate visual attributes into textual form, which is then externalized to NETWORK and STORAGE channels. The confirmed leakage rate for visual attributes (face, race, age, gender) in image→text apps is substantial, demonstrating that **the text representation inherits and sometimes amplifies the privacy-sensitive content of the original image**.

6.4 Attribute Persistence vs. Transformation

Not all attributes persist from input to externalization. Some (e.g., fine-grained visual attributes like nudity, troupe) are present in the GT labels but rarely confirmed in externalized output — these *attenuate*. Others (e.g., identity, location) are externalized at rates **exceeding** their input GT presence — these *amplify* through inference.

6.5 Modality Split Analysis (Docs / Image / Text)

- **Docs input:** input GT rate = 11.8%, confirmed leakage = 1.1%
- **Image input:** input GT rate = 35.2%, confirmed leakage = 2.1%
- **Text input:** input GT rate = 5.6%, confirmed leakage = 4.1%

Fig 12 — Input Type Comparison (Docs / Image / Text) × Attribute Family
(Docs = SROIE receipts; Image = HR-VISPR/MIMIC-CXR; Text = text datasets)

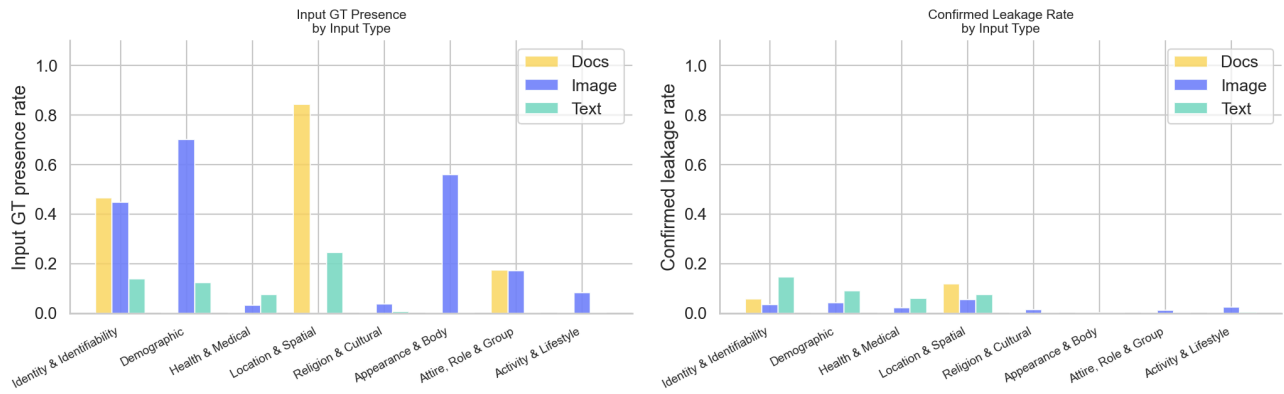
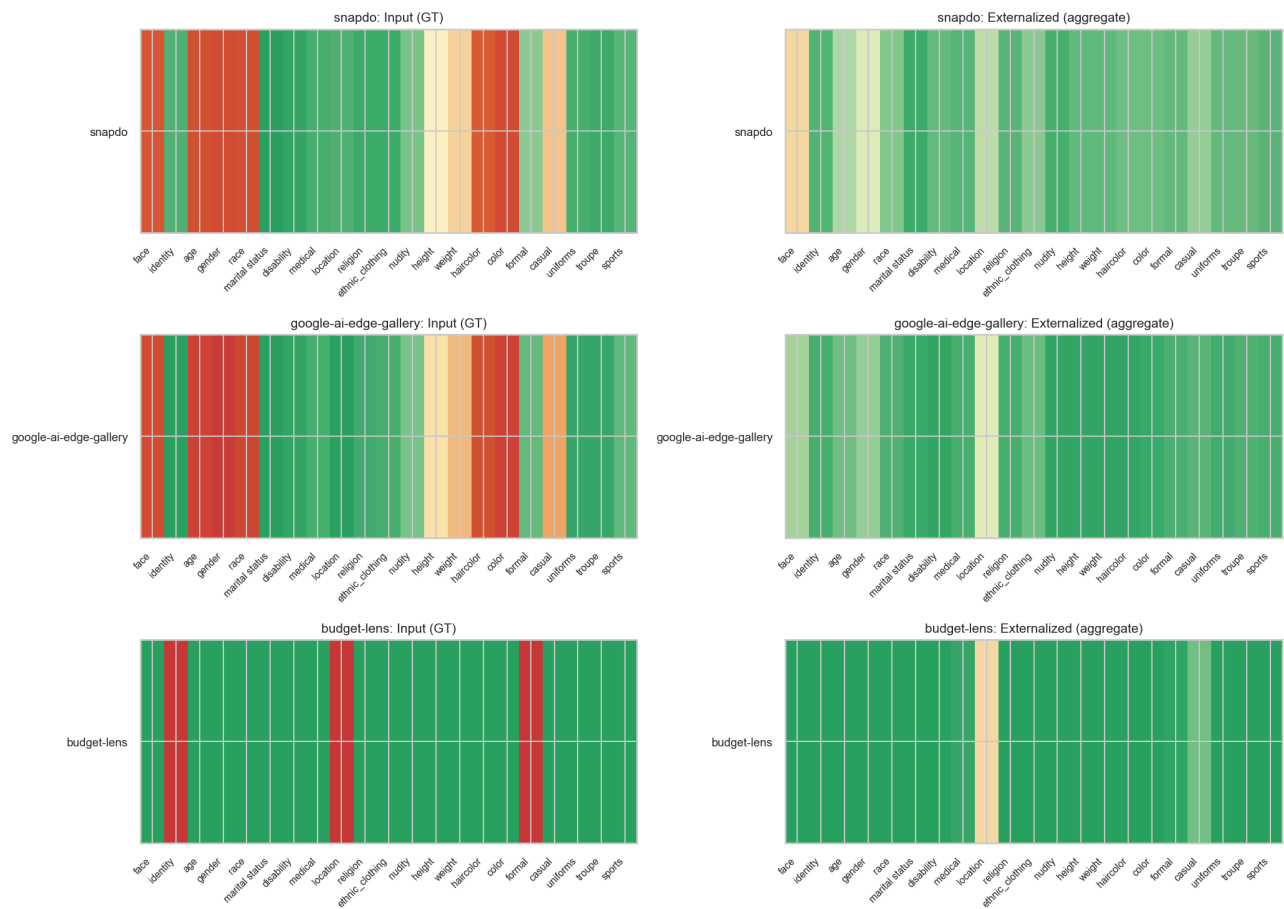


Fig 16 — Privacy Transformation Case Studies: Input GT vs Externalized Attribute Heatmaps
(0=no evidence; 1=possible; 2=confirmed; average across items)



6b. Leakage by App Category

Apps are grouped into six functional categories: **Finance**, **Photo/Camera**, **Productivity**, **Education**, **Social/Communication**, and **Health/Fitness**.

- **Health/Fitness** (257.0 items): 12.1% confirmed, 20.8% any leakage
- **Photo/Camera** (11.0 items): 9.1% confirmed, 27.7% any leakage
- **Education** (366.0 items): 3.4% confirmed, 11.1% any leakage
- **Social/Comm.** (430.0 items): 2.3% confirmed, 9.7% any leakage

- **Productivity** (1,156.0 items): 1.9% confirmed, 16.0% any leakage
- **Finance** (362.0 items): 1.1% confirmed, 6.7% any leakage

Fig 21 — Externalization Verdict Distribution by App Category

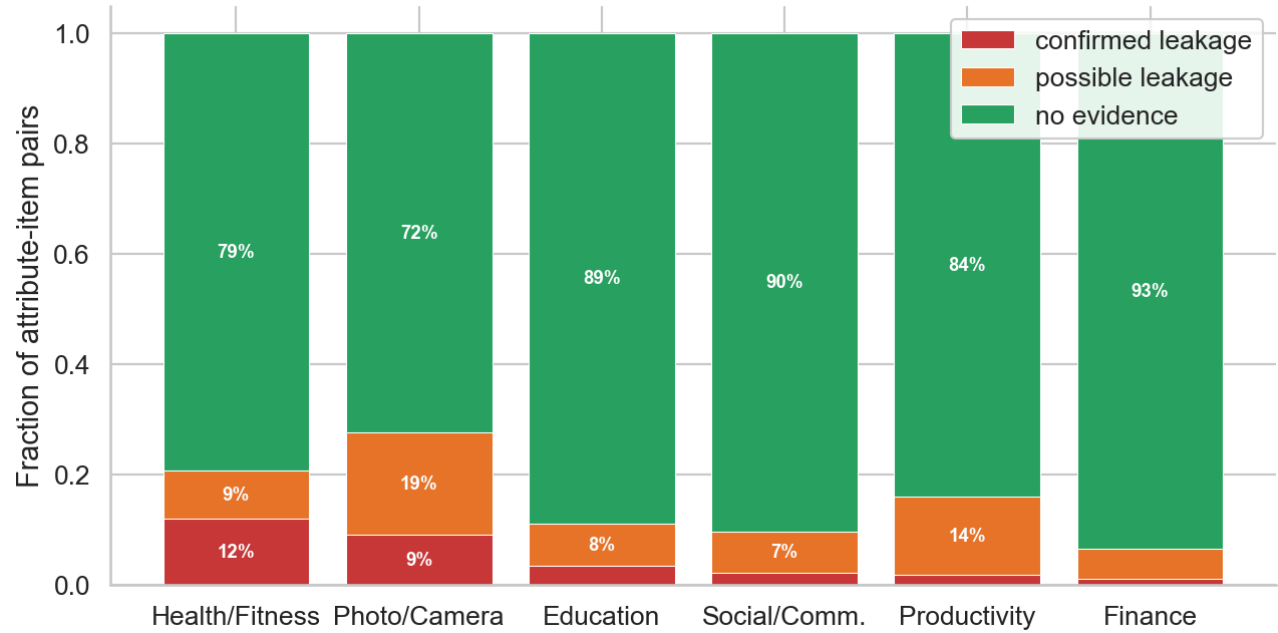
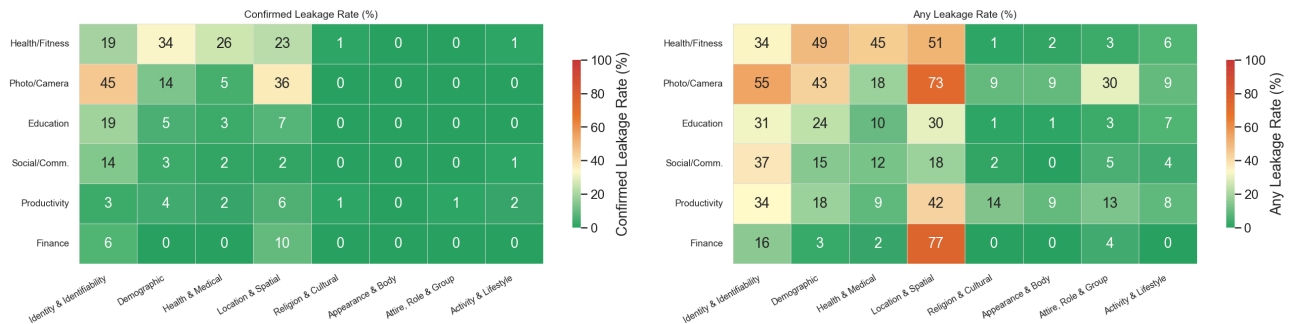
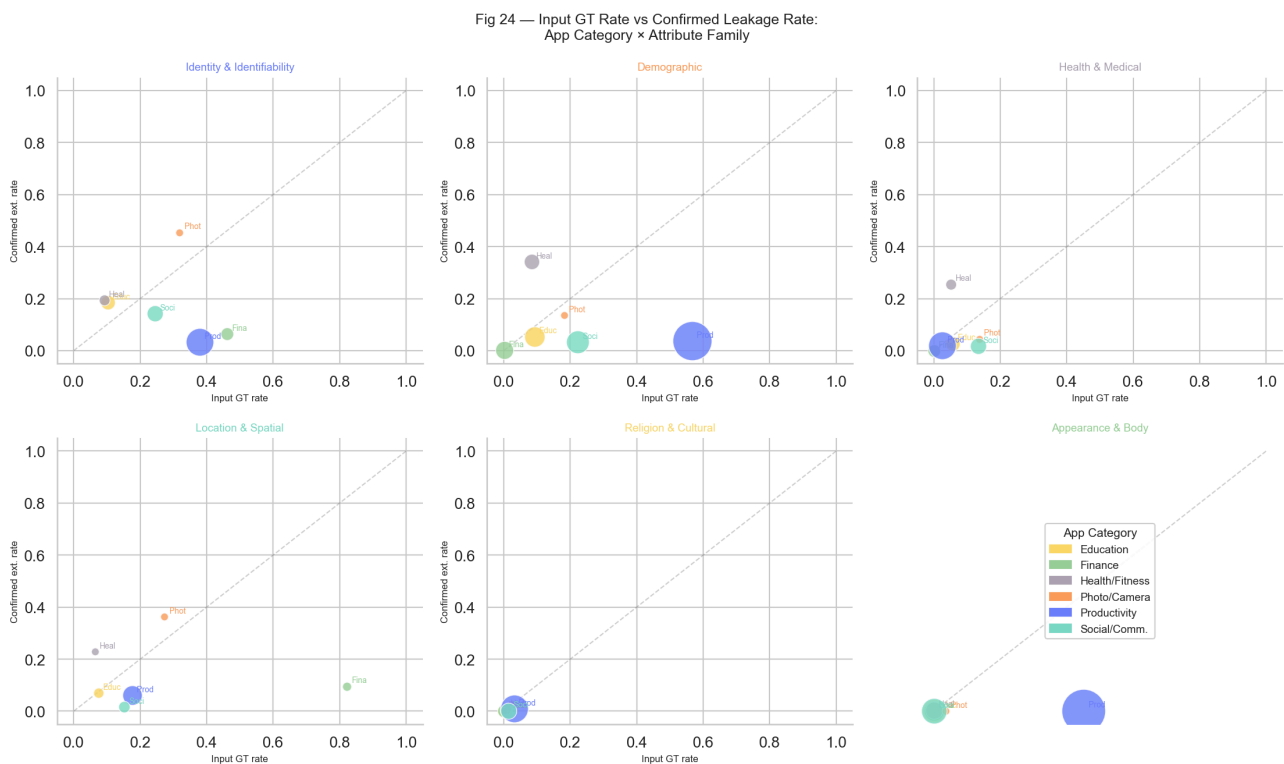
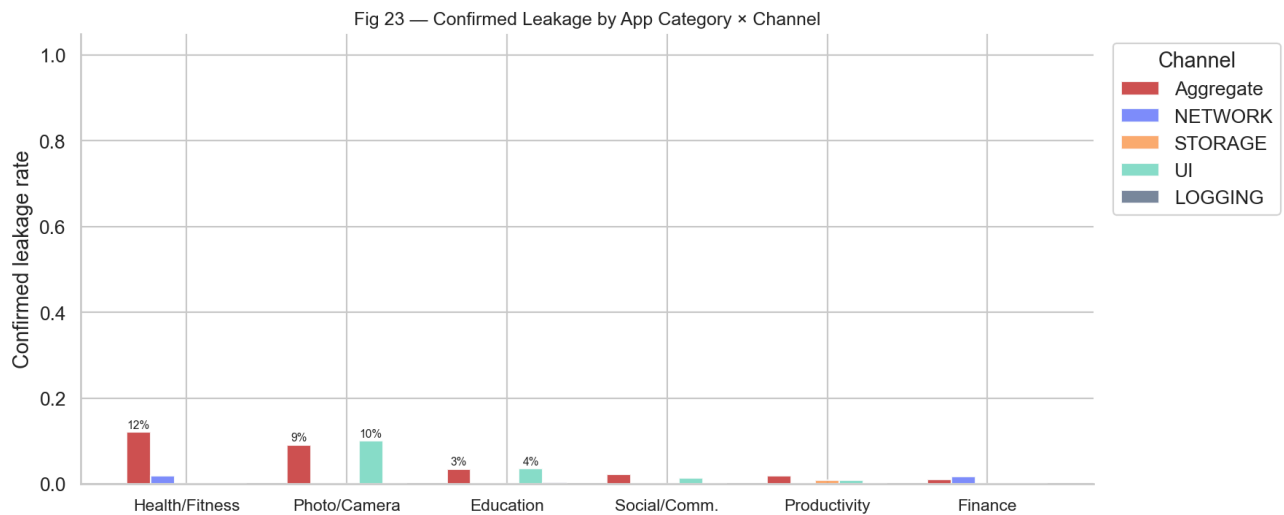


Fig 22 — Leakage Rate by App Category × Attribute Family





7. Recommendations

1. **Prioritize identity and location filtering** across all app types — these attributes leak at the highest rate regardless of input modality.
2. **Audit STORAGE channels** (databases, memory systems, on-device logs) — STORAGE leakage is non-visible to users and persists beyond session boundaries.
3. **Deploy inference-aware output filters for image→text apps** — the text description of an image is a semantic consolidation point for many visual attributes that were diffuse in the original image.
4. **Test for inference expansion** — apps should be evaluated not just for passthrough of GT-present attributes, but for attributes that appear in the output that were NOT in the input (implicit inference).

5. **Treat multi-attribute leakage as the norm** — privacy defenses that target individual attribute leakage miss the broader threat of simultaneous multi-attribute profile construction.

Appendix

A. Per-App Detailed Statistics

App	N pairs	Confirmed %	Any Leak %
healyks	3,465	13.1%	21.0%
waico	2,121	10.5%	20.6%
tool-neuron	231	9.1%	27.7%
llm-vtuber	3,969	6.1%	14.0%
deeptutor	7,812	4.3%	10.6%
chat-driven-expense-tracker	2,667	3.1%	9.3%
snapdo	17,808	2.5%	23.8%
google-ai-edge-gallery	8,505	1.9%	12.1%
pocketpal-ai	5,943	1.2%	8.8%
xend	8,421	0.5%	7.7%
edupal	2,079	0.1%	13.1%
budget-lens	4,956	0.0%	5.2%
clone	4,074	0.0%	0.3%
spendsense	42	0.0%	4.8%

B. Per-Dataset Sample Counts

Dataset	Input Type	N pairs	Confirmed %
ASAP-AES	text	5,817	0.1%
GretelSyntheticPII	docs	2,667	3.1%
HR-VISPR	image	29,148	2.0%
MIMIC-CXR	image	210	3.8%
MultiCaRe	text	3,675	13.3%
OpenPII	text	210	2.9%
PrivacyLens	text	19,005	4.5%
SROIE2019	docs	6,048	0.3%
SynthPAI	text	5,313	0.5%

C. Attribute Family → Member Mapping

Family	Attributes
Identity & Identifiability	face, identity

Demographic	age, gender, race, marital status
Health & Medical	disability, medical
Location & Spatial	location
Religion & Cultural	religion, ethnic_clothing
Appearance & Body	nudity, height, weight, haircolor, color
Attire, Role & Group	formal, casual, uniforms, troupe
Activity & Lifestyle	sports

D. Data Quality Notes

- prompt4/5 items with failed `ext_eval` (no verdict) are excluded from all analyses.
- Per-(app, dataset) groups with fewer than 5 verdict rows are excluded; only `tool-neuron|HR-VISPR` (1 item, image→image) and `spendsense|SR0IE2019` (2 items) remain in the very-small-sample regime — interpret their per-app rates with caution.
- `output_eval` (raw app output stage) is only available for PrivacyLens text→text runs (deeptutor, waico, llm-vtuber, tool-neuron, xend).
- Channel-level statistics are conditioned on the channel being present in that item's externalization record.